

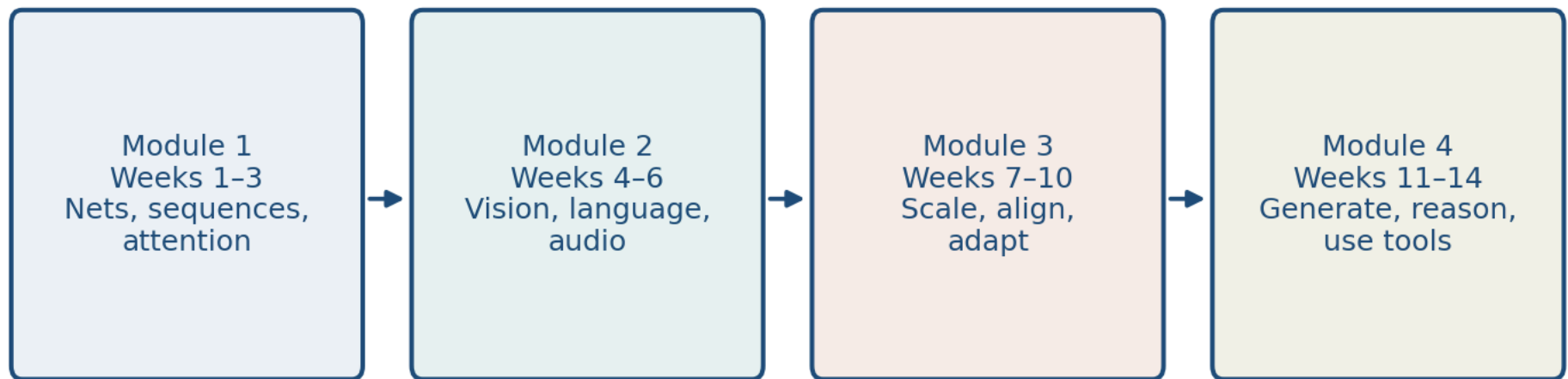
Learning goals for this block

- Place Week 1 at the base of an LLM stack.
- Write the next-token product and its negative log-likelihood.
- Compute the Llama 2 70B weights-file size (140 GB at float16).
- Turn “use an LLM” into a question plus a metric.

A self-contained LLM course

- DATA 441/641 and 442/642 are not prerequisites.
- Listed prereq: DATA 427/627.
- Overlap with those courses is intentional: more people can take this.
- The semester is a project. Lectures exist so that project has a backbone.

Where Week 1 sits: foundations before transformers



A language model is a stack

- Week 1 is the bottom: neurons, training, word vectors.
- Week 3 puts attention on those vectors.
- Later weeks align, retrieve, generate, and use tools.
- If your project “uses GPT,” name the layer you will change.

A language model is a stack. Week 1 is the bottom two layers.

Applications: RAG, tools, multimodal products

Alignment and adaptation: SFT, RLHF, LoRA, retrieval

Transformer: self-attention over token embeddings

Token embeddings: distributed word (and later patch) vectors

This week: neurons, activations, gradient descent

Pretraining minimizes next-token NLL

- $P(w_1, \dots, w_T) = \prod_t P(w_t \mid w_{<t})$.
- $L = - \sum_t \log P(w_t \mid w_{<t})$. Loss = empirical risk = cost.
- If $P(\text{mat} \mid \text{the cat sat on the}) = 0.70$, $\text{NLL} \approx 0.357$.
- XOR and skip-gram this week are the same kind of scalar, small.

Llama 2 70B is a 140 GB file

- 70×10^9 parameters $\times$ 2 bytes (float16) = 140 GB.
- Parameters file vs run file: running needs extra memory.
- Karpathy: an LLM is weights plus a little code.
- In class: 0:00–8:00. Finish 0:00–20:00 as homework.

The project starts this week

- A question you can measure, a public dataset, a baseline, one justified change.
- Bad: “Use LoRA on news.” Better: a metric on a 100-item holdout.
- Undergraduate: group report and talk. Graduate: main project plus a paper mini-project.
- Lab 1 ends with a five-line topic seed. You may change it.

Today's two-hour meeting

- 1.1 stack + project + short Karpathy (~26 min).
- 1.2 neuron / XOR (~16). 1.3 GD + one chain-rule (~18).
- 1.4 cosine + skip-gram softmax + bias (~18). Lab 1 start (~32).
- Books: Goodfellow 6.1, Nielsen Ch. 1, Jurafsky Ch. 5 (Aug 2026 draft), Mikolov §§1–3.